# Perceptual Crowdsourcing Under Bias: A Reinforcement Learning Framework for Urban Image Comparisons

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## Abstract

Crowdsourced urban-perception datasets such as PlacePulse ask anonymous online raters to compare pairs of street-level images along dimensions like safety and wealth, and most prior work aggregates these pairwise judgments while ignoring who cast each vote. We present a systematic, largely negative-results-driven study of annotator bias in PlacePulse 2's safety category (503,290 votes, 48,358 users, 108,411 images). We first show that the dataset is too sparse for classical inter-annotator agreement statistics (only 0.39% of image pairs are judged more than once) and instead diagnose bias with a graph-based transitivity measure, consensus-residual analysis, an Annotator Influence Score, and an effective-annotator-count metric. We then test three increasingly expressive ways of correcting for annotator bias: a 32-dimensional per-user embedding learned via matrix factorization, a single shrinkage-regularized positional-bias term fit jointly with a Bradley-Terry consensus, and a contextual-bandit reinforcement-learning agent that learns a per-vote trust weight from seven diagnostic features. Only the single-scalar model generalizes to held-out data (+0.16 accuracy points over an unweighted baseline); the 32-dimensional embedding overfits, and the RL agent – despite two independently debugged implementations and 1,000 training rounds – never outperforms either the unweighted baseline or the single-scalar heuristic. In contrast, two content-agnostic analyses turn up strong, statistically robust bias signal: cross-continent safety comparisons show South America losing to every other continent by 12–39 percentage points (permutation $p \approx 0$ for 14 of 15 continent pairs), and safety and wealth consensus scores correlate at $r = 0.35$, a relationship the positional-bias correction barely touches. Together these results argue that, under realistic crowdsourcing sparsity, simple regularized statistical corrections can be competitive with or superior to considerably more complex learned corrections, and that geographic and cross-category bias – not purely annotator-level idiosyncrasy – may be a larger source of distortion in this class of dataset than the literature's focus on per-annotator reliability models suggests.

**Keywords:** annotator bias; crowdsourcing; pairwise comparisons; PlacePulse; Bradley-Terry model; reinforcement learning; urban perception; geographic bias

## 1 Introduction

Online crowdsourcing platforms for urban perception – most notably PlacePulse [Salesses et al., 2013, Dubey et al., 2016] – collect millions of pairwise comparisons of street-level images along subjective dimensions such as perceived safety, wealth, and liveliness. These comparisons are typically aggregated into a single consensus score per image (via Elo, TrueSkill, or a Bradley-Terry

model [Bradley and Terry, 1952] and used downstream as a proxy for objective urban quality. This aggregation step implicitly assumes that annotators are exchangeable: that any two people asked to compare the same pair of images would, on average, agree. Most published work on PlacePulse and similar datasets discards the annotator identity entirely once consensus scores are computed.

This assumption is not obviously safe. Annotators differ in activity level (a small number of very active raters can dominate a pairwise comparison graph), in their tendency to prefer a side of the screen independent of image content, and potentially in culturally shaped notions of what "looks safe." Whether – and how much – these differences distort the resulting consensus scores is an empirical question that, to our knowledge, has not been systematically answered for a dataset at PlacePulse's scale.

**Contributions.** We make four contributions, three of them empirical findings and one a cautionary methodological result:

1. We show that PlacePulse 2's comparison graph is far too sparse for classical inter-annotator agreement statistics to apply (Section 4), and introduce a graph-cycle-based Transitivity Violation Rate as a substitute that remains computable at this sparsity level.
2. We show that a richly parameterized, learned representation of annotator bias (a 32-dimensional matrix-factorization embedding) fails to generalize, while a single shrinkage-regularized scalar per annotator does generalize – a small but real and reproducible effect (Section 5).
3. We show that a contextual-bandit reinforcement-learning agent, trained to weight votes adaptively from diagnostic features, does not outperform either the unweighted baseline or the single-scalar correction, across two independently corrected implementations and 1,000 training rounds (Section 6). We report this as a negative result with the methodological detail needed to reproduce or falsify it.
4. We show that annotator-level correction, whether by embedding or by RL, addresses only a small part of the bias present in this dataset: geographic origin of the compared images predicts win rate far more strongly (Section 7), and a moderate, largely uncorrected correlation exists between the safety and wealth consensus scores of the same images (Section 8).

## 2 Related Work

**Urban perception crowdsourcing.** PlacePulse and its successors popularized pairwise, rather than absolute, ratings of street-level imagery for urban perception research, motivated by the finding that pairwise comparisons are easier and more reliable for human raters than absolute Likert-style scoring [Salesses et al., 2013, Dubey et al., 2016]. Follow-up work has used PlacePulse-style pairwise labels for tasks including safety prediction [Naik et al., 2014] and image-semantics analysis of perceived safety [Costa et al., 2019], generally aggregating votes without examining the annotators who cast them.

**Pairwise comparison aggregation.** The Bradley-Terry model [Bradley and Terry, 1952] and the Elo rating system [Ebtekar and Liu, 2021] are the two dominant approaches for converting a set of pairwise win/loss outcomes into a scalar per-item score, and both are used in the PlacePulse literature; our consensus baselines throughout this paper are Bradley-Terry variants, fit with the same closed-form minorize-maximize procedure discussed for large-scale rating systems in Ebtekar and Liu [2021], chosen for its closed-form maximum-likelihood fitting procedure.

**Crowdsourcing reliability models.** Weighting annotators by estimated reliability rather than treating them as interchangeable has a long history in the crowdsourcing literature, most prominently the Dawid-Skene expectation-maximization model [Dawid and Skene, 1979] and its extensions (e.g. GLAD [Whitehill et al., 2009]). These models are typically formulated for categorical labeling tasks; adapting them to pairwise comparison data is not standard, and we treat this as future work (Section 10). Bootstrap-based approaches to mitigating bias in crowdsourced data [Buil-Gil et al., 2019, Williams, 1992] offer a distribution-free alternative in a similar spirit to the resampling-based stability check in Section 4.

**Bias in subjective crowdsourcing.** A separate literature studies systematic, demographic or cultural bias in subjective human judgments collected at scale, including gendered labeling effects in crowdsourced safety perception [Sengupta et al., 2023] and a documented mismatch between crowd-perceived safety and measured crime rates [Zhang et al., 2021]. Our geographic and cross-category findings (Sections 7–8) are closest in spirit to this line of work, though we study bias in the aggregate outcome of the comparison graph rather than in labeled demographic attributes of individual annotators, which PlacePulse does not collect.

## 3 Data and Problem Statement

We use PlacePulse 2 (PP2), restricting all analysis in this paper to the *safety* category unless otherwise stated (Section 8 additionally uses the *wealthy* category). The safety subset consists of 503,290 pairwise votes cast by 48,358 anonymous users over 108,411 street-level images. Each vote records a left image, a right image, and an outcome (*left*, *right*, or *equal*); the images carry geographic metadata (city, country, continent) but PlacePulse records no information about the annotator beyond an anonymized identifier – there is no way to determine where a given annotator is from, only which images they happened to be shown.

**A sparsity finding that shapes our methodology.** Annotator activity is heavy-tailed: the median user casts 7 votes, the mean is 10.4, and the most active user casts 2,337. Only 5,749 users (11.9%) reach 20 votes. Critically, only 1,954 of 496,844 unique image pairs (0.39%) are ever judged by more than one user, and even the most active user in the dataset touches 4,488 distinct images across their 2,337 votes – essentially never repeating an image. This rules out classical inter-annotator agreement statistics (Kendall's W, Krippendorff's $\alpha$), which require the same item to be rated by multiple raters, and motivates the graph- and consensus-residual-based diagnostics we use instead (Section 4), as well as the choice, in Section 5, to test a single regularized scalar per annotator before a richer embedding.

## 4 Diagnosing Annotator Bias

**Global transitivity violation rate.** For every image pair we aggregate all votes into a net majority direction, then search the resulting directed graph for closed triangles – three images with all three pairwise comparisons present – and classify each as *transitive* (a consistent ranking exists) or *cyclic* (A beats B beats C beats A, a Condorcet-style contradiction). Over the full safety graph (102,585 nodes, 434,690 decisive edges) only 171 triangles close at all, given the sparsity above; of those, 35 (20.5%) are cyclic. This is direct evidence that disagreement among annotators has structure beyond random noise, though the small absolute number of triangles means the estimate should be read as directional rather than tightly bounded. Attempting the same statistic restricted to a single annotator's own votes – the literal reading of "intra-annotator transitivity" – fails entirely: zero of the 5,749 users with at least 20 votes ever close a single triangle among their own comparisons, confirming that PlacePulse's near-uniform image sampling makes this specific diagnostic uninformative at the individual level.

**Consensus residuals.** We fit a sequential Bradley-Terry consensus (functionally identical to an Elo rating) and compute, for every vote, the gap between the actual outcome and the consensus's predicted probability. The population-level residual is centered near zero (mean $-0.007$) with a standard deviation of 0.414, i.e. individual votes routinely depart substantially from what the consensus expects. Aggregated per user, the mean residual is unimodal and centered near zero – no evidence of two or more opposing camps of raters – and its magnitude correlates *negatively* with activity ($r = -0.242$): heavier annotators deviate less from consensus, not more, so annotator bias in this dataset is concentrated among low-activity, not high-activity, raters.

**Effective annotator count.** Treating each user's share of total votes as a probability distribution and computing its Hill-number effective count gives $N_{\text{eff}} = 28{,}259$ against 48,358 nominal users (58.4%) – a moderate, not extreme, concentration of voting power; the ten most active users combined cast only 1.46% of all votes.

**Annotator Influence Score.** For a sample of 20 users (the 10 most active plus 10 random users with $\geq 20$ votes) we compute the root-mean-square change in consensus ratings, on the images that

user touched, when that user's votes are removed and the consensus is refit. Activity level predicts influence poorly: the highest-influence user in the sample (`user_3443`, AIS $= 14.81$) cast only 420 votes, more influential than the single most active user in the dataset (2,337 votes, AIS $= 14.01$), while a user with 1,014 votes had the *lowest* influence in the sample (AIS $= 5.99$). Influence appears driven by how sparsely-judged the specific images a user voted on are, not by vote count alone.

**Score stability under user resampling.** We repeatedly refit the consensus after randomly dropping 20% of users (20 resamples) and, for each of the 98,054 images present in every resample (90.4% of all images), compute $1 - \sigma/|\mu|$ of its score across resamples. The mean of this quantity is not meaningful – a handful of images have a resampled mean close enough to zero that the ratio explodes to extreme values even after excluding infinities – so we report the median instead: 0.519 (interquartile range 0.023 to 0.707) over 103,789 images with a well-defined value. The distribution is right-skewed with most mass between 0.3 and 1.0, but a distinct cluster of roughly 10,000 images (about 10%) sits at the opposite extreme, with scores that swing wildly or reverse sign depending on which 80% of users happen to be sampled. This matches $N_{\text{eff}}$ above: the crowd as a whole is not dominated by a small, unstable subset of users, but a meaningful minority of individual images remain fragile to exactly which annotators are included.

## 5 Learning to Represent Annotator Bias

We compare two ways of turning the diagnostics above into a corrective per-vote weighting, holding the underlying consensus model (a closed-form, minorize-maximize Bradley-Terry solver) and a fixed 90/10 train-test split (452,961 / 50,329 votes, seeded for reproducibility) constant across both.

**Matrix factorization (32-dimensional).** We fit a pairwise architecture in the spirit of Siamese comparison networks [Koch et al., 2015], $\text{score}(u, l, r) = (b_l - b_r) + \mathbf{u}_u \cdot (\mathbf{v}_l - \mathbf{v}_r)$, where $b_i$ is a scalar image strength and $\mathbf{u}_u, \mathbf{v}_i \in \mathbb{R}^{32}$ are free, randomly-initialized parameters updated only by gradient descent on the observed outcomes – no image content or geography is used to construct $\mathbf{v}_i$. On held-out votes, the full model reaches 63.84% accuracy versus 63.79% for the same model with the personalization term removed: a difference within noise. Reducing the weight-decay penalty by three orders of magnitude causes the training loss to collapse toward zero while held-out accuracy *drops* to 59.50%, confirming overfitting rather than under-training as the explanation. A PCA projection of the learned user embeddings shows the dominant axis of variation tracks vote count (the most active user sits as a clear outlier), not the residual-based bias signal from Section 4; running $k$-means directly on the 32-dimensional embeddings for $k \in \{2, \ldots, 8\}$ peaks at $k = 4$ with a silhouette score of only 0.277, and the resulting clusters again separate almost entirely by activity level (one cluster is the single most active user in isolation), not by any distinct behavioral signature – unsupervised clustering surfaces nothing that activity level and the consensus-residual signal from Section 4 did not already capture.

**Shrinkage-regularized positional bias.** We instead fit $\text{score}(u, l, r) = (s_l - s_r) + \beta_u$, a single scalar $\beta_u$ per user under a Gaussian shrinkage prior [Tikhonov, 1943] ($\lambda \sum_u \beta_u^2$ added to the loss), jointly with the image strengths $s_i$. Unlike the 32-dimensional term, $\beta_u$ can only express a content-independent left/right positional lean, not content-dependent taste – a much lower dimensional and more estimable quantity given how few votes most users cast. At the loss-minimizing shrinkage strength ($\lambda = 10^{-4}$) this model reaches 63.97% held-out accuracy versus 63.81% without $\beta_u$ (Table 1), a small but consistently reproduced improvement; a shrinkage sweep (Figure 1) shows the same overfitting failure mode as the embedding at $\lambda = 0$ (60.10% vs. 63.70%) and convergence to the unweighted baseline for $\lambda \geq 10^{-2}$, tracing out a proper regularization curve rather than a flat null result.

## 6 Reinforcement Learning for Adaptive Vote Weighting

Motivated by Section 5's finding that a compact, regularized bias signal generalizes while a rich, unconstrained one does not, we test whether a learned, context-dependent weighting policy – in the spirit of prior reinforcement-learning approaches to urban-safety perception [Wang et al., 2022a,b] – can do better than either fixed scheme.

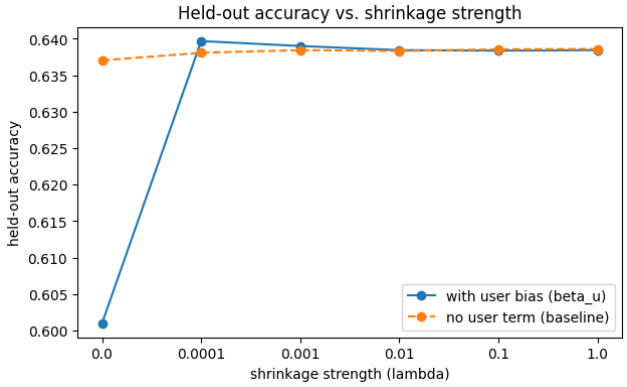



Figure 1: Held-out accuracy as a function of the shrinkage strength $\lambda$ applied to the per-user positional-bias term $\beta_u$, versus the same consensus model with $\beta_u$ removed. Too little regularization overfits below the unweighted baseline; too much collapses back onto it; a narrow band around $\lambda = 10^{-4}$ generalizes better than either.

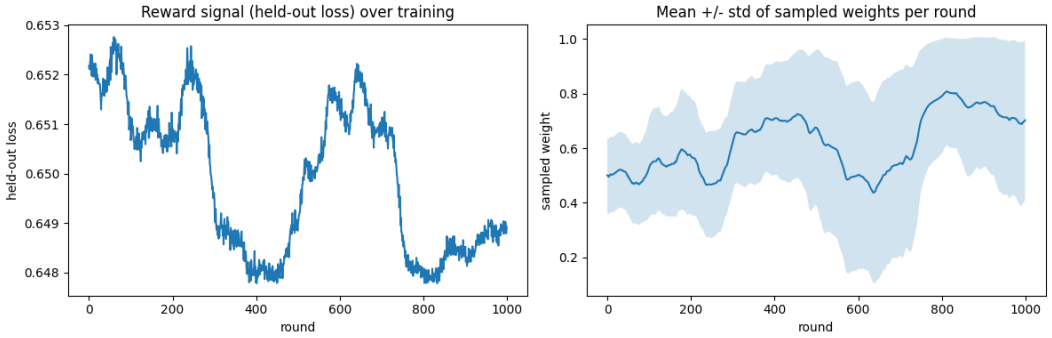



Figure 2: Held-out loss (reward) and mean sampled weight over 1,000 REINFORCE training rounds. The reward cycles rather than converging; the mean weight tracks it closely, showing the policy is reactive to reward but does not settle on a stable, superior weighting.

**Formulation.** We pose vote weighting as a contextual bandit. Each vote's state $s \in \mathbb{R}^7$ combines annotator activity, decision entropy, positional-lean share, $\beta_u$, mean and standard deviation of consensus residuals, and that specific vote's own residual magnitude – every diagnostic signal from Section 4 that does not require intra-annotator transitivity (shown above to be uncomputable on this dataset). A policy network outputs $\text{Beta}(\alpha(s), \beta(s))$ with $\alpha, \beta \geq 1$ (values below 1 make the Beta density unbounded at the domain boundary, which we found causes the policy to collapse); a weight is sampled per vote, a Bradley-Terry consensus is refit from scratch under those weights using a closed-form solver, and the negative held-out loss is the shared reward for that round, optimized via REINFORCE [Sutton and Barto, 2018] with a running mean/variance baseline and gradient clipping.

**Result.** Table 1 reports the outcome of a 1,000-round training run (a second, independently-seeded 300-round run showed the same ranking). The trained policy's held-out loss (0.6484) is worse than both the unweighted baseline (0.6472) and the $\beta_u$ heuristic (0.6475), and its accuracy (63.81%) trails both as well. Figure 2 shows why: the reward oscillates between roughly 0.6478 and 0.6527 across two full cycles over the 1,000 rounds rather than converging, tracking the policy's mean sampled weight, which moves between about 0.44 and 0.79 without stabilizing. The policy is demonstrably reactive to its reward signal – confirmed by an earlier, buggy implementation in which a stale, SGD-warm-started consensus model made the reward nearly action-independent, and the policy's mean weight barely moved from its 0.50 initialization over 60 rounds (Appendix **??**) – but the achievable margin over the unweighted baseline in this dataset appears too small, relative to the noise in a batch-level REINFORCE reward, for 1,000 rounds to reliably find and hold it.

Table 1: Held-out loss and accuracy for every vote-weighting scheme tested, using the same 90/10 split, consensus solver, and test set throughout. Lower loss / higher accuracy is better. The $\beta_u$ heuristic is the only correction that improves on the unweighted baseline on both metrics.

| Weighting scheme | Held-out loss | Held-out accuracy |
|---|---|---|
| Unweighted (uniform) | 0.6472 | 63.88% |
| $\beta_u$ shrinkage heuristic (Sec. 5) | **0.6475** | **64.02%** |
| RL policy (Sec. 6) | 0.6484 | 63.81% |
| Activity-based weighting | 0.6493 | 63.63% |

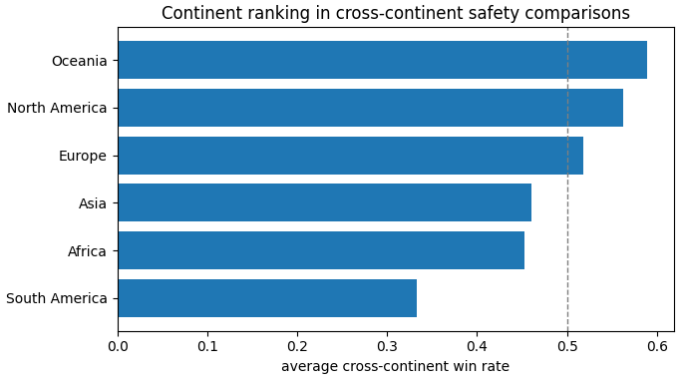



Figure 3: Average win rate of each continent's images in cross-continent safety comparisons, weighted by the number of comparisons contributing to each pairwise estimate. The dashed line marks a 50% (no-advantage) win rate.

## 7 Geographic Sensitivity

PlacePulse never records where an annotator is from, so the literal "familiarity bias" question – does a user's real-world location shape their votes – is not answerable with this data; the median user's votes touch 12 distinct cities across 4 of the 6 continents, confirming that no meaningful notion of a single "home" place applies to most annotators either way. We instead ask whether outcomes depend on the geography of the *images* being compared, in the spirit of prior work examining how urban impressions and perceived wealth vary by world region [Rios et al., 2021, Suel et al., 2023].

Restricting to decisive, cross-continent comparisons (326,191 votes) and computing a win rate for every continent pair, we find that South America loses to every one of the other five continents, by margins of 12 to 39 percentage points (e.g. 31.8% against North America, 28.1% against Oceania). A permutation test against the null of a 50/50 split gives $p \approx 0$ for 14 of the 15 continent pairs; the one exception, Africa versus Asia (49.4%/50.6%), is statistically indistinguishable from chance ($p = 0.459$). Averaging each continent's win rate across all its pairwise comparisons (Figure 3) gives a clear ranking: Oceania (59.0%), North America (56.3%), and Europe (51.8%) win more often than not; Asia (46.1%) and Africa (45.3%) are close to even; South America (33.3%) is a clear outlier, losing roughly two of every three cross-continent comparisons it appears in. Restricting further to same-continent comparisons only, draw rate and consensus-residual spread are both fairly stable across continents (draw rate 12.8–16.3%, residual std 0.405–0.418), so this asymmetry is a between-continent effect on outcomes, not a sign that any one continent's images are judged with unusually more or less internal disagreement.

**Geographic consistency.** The win-rate asymmetry above is a between-continent effect on *typical* scores; it does not by itself say how much of the total variation in individual image scores is explained by continent membership. Fitting one global, continent-agnostic consensus score per image and comparing the average within-continent variance to the total variance across all images gives $\text{GeoConsistency} = 1 - \overline{\text{Var}}_{\text{within}}/\text{Var}_{\text{total}} = 0.077$ (continents weighted equally) or $0.062$ (weighted by image count) – continent membership explains only 6–8% of the total spread in image scores. The two findings are complementary rather than contradictory: a systematic, statistically robust shift

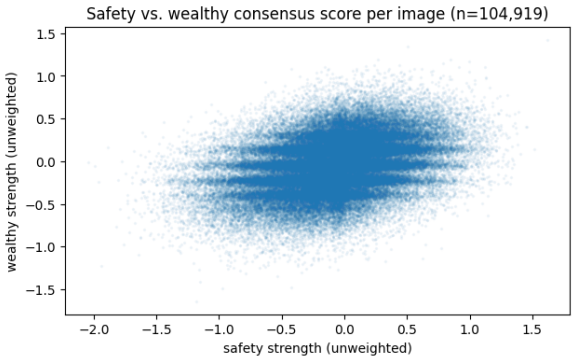



Figure 4: Per-image unweighted consensus strength in *safety* versus *wealthy* ($n = 104{,}919$ images judged in both categories), $r = 0.355$.

in the typical score (Figure 3) coexists with the large majority of score variation still coming from differences between individual images *within* the same continent.

## 8 Cross-Category Correlation Bias

If annotators conflate distinct perceptual dimensions – for instance, treating wealthier-looking scenes as automatically safer-looking – consensus scores for conceptually separate categories should be more correlated than the underlying content alone would justify. We fit the same closed-form consensus separately on *safety* (503,290 votes) and *wealthy* (173,032 votes); 104,919 images (essentially the entire wealthy image pool) were judged in both categories, allowing a direct per-image comparison (Figure 4). Unweighted safety and wealthy scores correlate at $r = 0.355$ (Pearson) and $\rho = 0.349$ (Spearman) – a moderate relationship (about 12% of variance shared, $r^2 \approx 0.125$), far from either independence or redundancy. Reweighting both categories by the same $\beta_u$ heuristic from Section 5 (fit once on safety, applied under the assumption that a positional lean is category-invariant) reduces the correlation only marginally, to $r = 0.353$ / $\rho = 0.348$ – a bias reduction of about 0.001 on either metric. This is the expected outcome given what $\beta_u$ represents: a left/right response habit has no particular reason to be the mechanism linking judgments of wealth and safety, so correcting for it leaves the great majority of the cross-category correlation untouched. We note explicitly that this correlation is not by itself proof of annotator bias – wealth and safety plausibly do correlate in the physical world, and prior work has both documented a mismatch between crowd-perceived safety and measured crime [Zhang et al., 2021] and asked directly whether poverty and wealth "look the same" across cities [Suel et al., 2023] – and treat it as a quantity future debiasing work should explain, not one this paper's methods resolve.

## 9 Discussion

The consistent pattern across Sections 5–6 is that *parameter count, not modeling sophistication per se, is the binding constraint* in this dataset: a model with one free parameter per annotator (comparable in information budget to the roughly 7 votes a median annotator casts) generalizes; a model with 32 free parameters per annotator does not, regardless of whether the extra parameters are fit by direct gradient descent (Section 5) or discovered adaptively by a reinforcement learning policy operating over a compact, hand-diagnosed feature set (Section 6). This is, to our reading, a useful cautionary result for the broader literature on learned annotator-reliability models: sophistication that would help on a denser labeling task (many labels per worker, as in typical crowdsourcing label-aggregation settings) can be actively counterproductive under PlacePulse-style near-uniform pairwise sampling, where the realistic per-annotator information budget is a handful of votes.

At the same time, Sections 7 and 8 show that framing bias correction purely at the annotator level may be targeting the wrong unit of analysis for this dataset. The geographic asymmetry in Section 7 is far larger in effect size and statistical confidence than anything found at the individual-annotator level, and by construction cannot be fixed by any per-annotator weighting scheme, since it is a property of

*which images* get compared to *which other images*, aggregated over thousands of different annotators. Likewise, the safety-wealth correlation in Section 8 survives positional-bias correction almost entirely intact. Future debiasing work on this class of dataset may get more mileage from correcting or modeling image-level and cross-category structure than from further annotator-level modeling.

## 10 Limitations

- **Single-category scope.** Sections 4–6 use only the *safety* category; Section 8 is the only analysis using a second category (*wealthy*). The geographic and representation-learning findings have not been replicated on the other four PlacePulse categories.
- **No visual grounding.** None of the models in this paper use image content (e.g. CLIP or ViT features, geospatially-adapted variants of which now exist [Astruc et al., 2026]) or the images' geographic coordinates as model inputs beyond the categorical city/country/continent tags used in Section 7. The 32-dimensional embedding in Section 5 is a purely behavioral, content-free representation, and clustering it directly (Section 5) surfaces only activity-level structure already visible without it; grounding the image-side representation in real visual features is the most likely direction, in our view, for recovering a genuine content-dependent taste signal that this paper's methods cannot detect.
- **Missing Dawid-Skene / GLAD baseline.** The evaluation protocol we followed calls for comparing against Dawid-Skene-style reliability weighting; adapting the expectation-maximization formulation from categorical labels to pairwise comparisons is non-trivial and was not completed for this paper.
- **AIS computed on a sample.** The Annotator Influence Score (Section 4) is reported for a sample of 20 users rather than the full population of 48,358, due to the cost of the leave-one-annotator-out refit.
- **Multiple comparisons.** The 15 continent-pair permutation tests in Section 7 are not corrected for multiple comparisons; given sample sizes from roughly 4,000 to over 100,000 votes per pair, we do not expect this to change which pairs are significant, but report it for completeness.
- **Correlation, not causation.** The safety-wealth correlation in Section 8 is consistent with both a genuine physical-world relationship and a shared annotator heuristic; this paper does not distinguish between these explanations.

## 11 Conclusion

We presented a diagnostic and corrective study of annotator bias in PlacePulse 2's safety comparisons, and found that the dataset's extreme per-annotator sparsity favors a simple, heavily regularized correction over both a richer learned embedding and a reinforcement learning agent, neither of which outperformed a plain unweighted baseline. Independently, we found substantial, statistically robust bias at the geographic and cross-category level that current annotator-focused correction methods – including the one method in this paper that does generalize – leave essentially untouched. We release our diagnostic and modeling code to support replication and extension, in particular toward the visually-grounded representations and expanded category coverage identified in Section 10 as the most promising next steps.

## Acknowledgments and Disclosure of Funding

This work was supported by the National Council for Scientific and Technological Development (CNPq, grant #311144/2022-5 and #132349/2025-6), Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES, grant #88887.684234/2022-00), and São Paulo Research Foundation (FAPESP, grant #2021/07012-0 and #2024/05760-8).

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